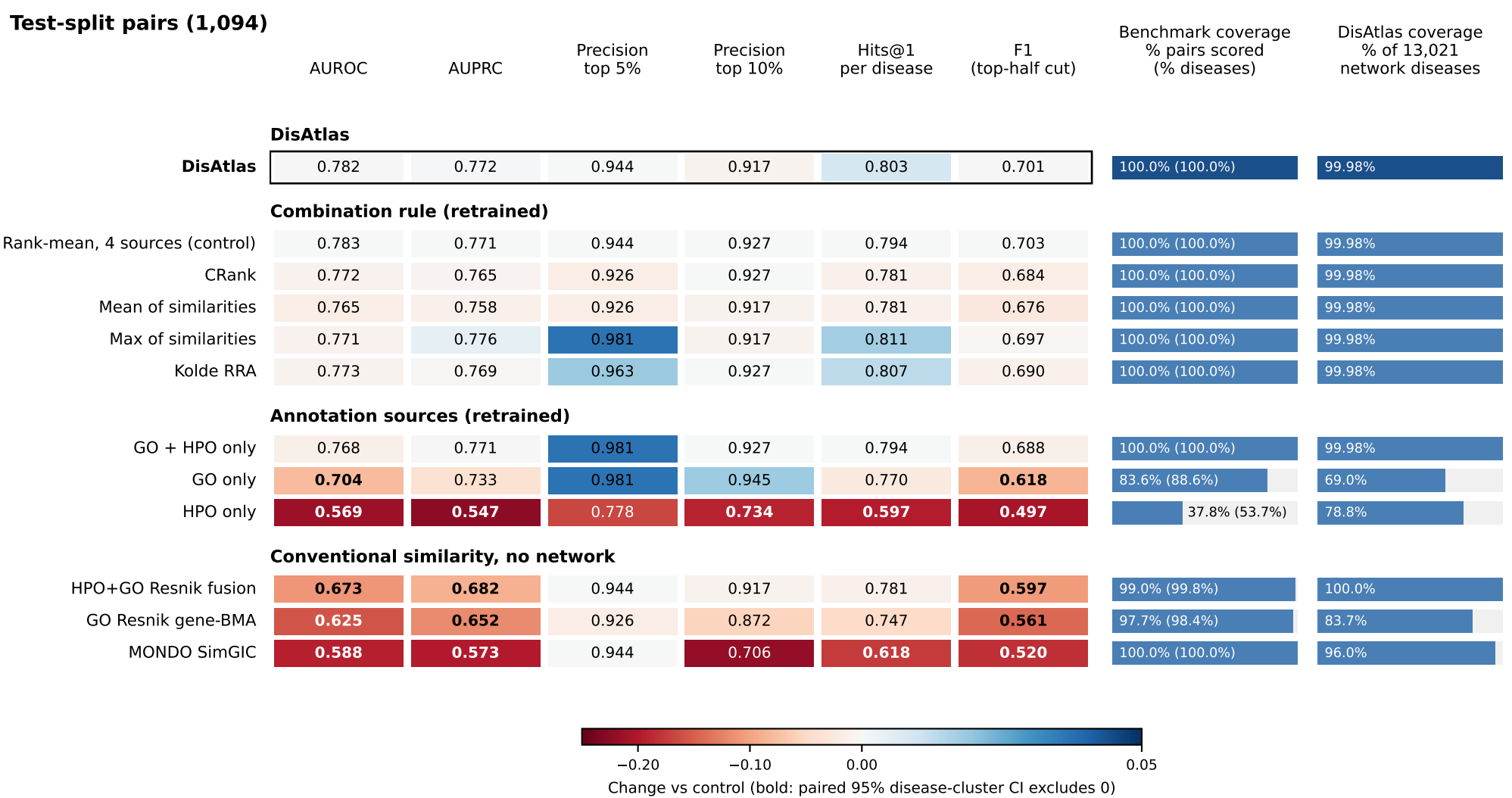
Test-split pairs (1,094)
AUROC
AUPRC
Precision top 5%
Precision top 10%
Hits@1 per disease
F1 (top-half cut)
Benchmark coverage % pairs scored (% diseases)
DisAtlas coverage % of 13,021 network diseases
DisAtlas
DisAtlas 0.782 0.772 0.944 0.917 0.803 0.701 100.0% (100.0%) 99.98%
Combination rule (retrained)
Rank-mean, 4 sources (control) 0.783 0.771 0.944 0.927 0.794 0.703 100.0% (100.0%) 99.98%
CRank 0.772 0.765 0.926 0.927 0.781 0.684 100.0% (100.0%) 99.98%
Mean of similarities 0.765 0.758 0.926 0.917 0.781 0.676 100.0% (100.0%) 99.98%
Max of similarities 0.771 0.776 0.981 0.917 0.811 0.697 100.0% (100.0%) 99.98%
Kolde RRA 0.773 0.769 0.963 0.927 0.807 0.690 100.0% (100.0%) 99.98%
Annotation sources (retrained)
GO + HPO only 0.768 0.771 0.981 0.927 0.794 0.688 100.0% (100.0%) 99.98%
GO only 0.704 0.733 0.981 0.945 0.770 0.618 83.6% (88.6%) 69.0%
HPO only 0.569 0.547 0.778 0.734 0.597 0.497 37.8% (53.7%) 78.8%
Conventional similarity, no network
HPO+GO Resnik fusion 0.673 0.682 0.944 0.917 0.781 0.597 99.0% (99.8%) 100.0%
GO Resnik gene-BMA 0.625 0.652 0.926 0.872 0.747 0.561 97.7% (98.4%) 83.7%
MONDO SimGIC 0.588 0.573 0.944 0.706 0.618 0.520 100.0% (100.0%) 96.0%
−0.20
−0.10
0.00
0.05
Change vs control (bold: paired 95% disease-cluster CI excludes 0)